Thermal Check Report (Power Tree – Actual Load Budget)

System regulators: LM2596 (9V→5V) + AMS1117-3.3V (5V→3.3V)

1. Inputs from Power Tree

Ambient temperature cases: Ta = 25°C (typ) and 60°C (worst enclosed).
LM2596 efficiency assumption: η = 80–85% (depends on module/layout and load).
Thermal resistance (θJA) estimates for quick check (PCB/module dependent):
• LM2596: 40–80 °C/W • AMS1117 (SOT-223): 50–100 °C/W

2. Load Summary

2.1 5V Rail Loads (from LM2596)

5V Load	Iavg (mA)	Ipeak (mA)
AMS1117 input (5V→3.3V)	35–40	80 (budget)
LCD with I2C backpack	25–35	90
MQ-2 gas sensor	180	250
HC-05 Bluetooth	40	80
Buzzer (ON)	40	100

5V rail totals (actual budget):

• Iavg (Buzzer OFF): 280–295 mA ⇒ Pout = 1.40–1.48 W

• Iavg (Buzzer ON): 320–335 mA ⇒ Pout = 1.60–1.68 W

• Ipeak (sum peaks): 600 mA ⇒ Pout = 3.00 W

2.2 3.3V Rail Loads (from AMS1117-3.3V)

3.3V Load	Iavg (mA)	Ipeak (mA)
STM32F103C8T6 (Bluepill)	35	70

DHT11	1	3

3.3V rail totals: Iavg ≈ 36 mA, Ipeak ≈ 73 mA (budgeted to 80 mA).

3. Thermal Check (Quick)

3.1 LM2596 (Buck) – 9V→5V

Formulas: Pout = 5V × Iout; Ploss ≈ Pout × (1/η − 1); ΔT ≈ Ploss × θJA; Tj ≈ Ta + ΔT

Case	Iout	Pout	Ploss (η=80–85%)	ΔT est. (θJA 40–80)	Tj est. @ Ta=25/60
5V average (Buzzer OFF)	280–295 mA	1.40–1.47 W	0.25–0.37 W	10–30 °C	25°C: 35–55°C 60°C: 70–90°C
5V average (Buzzer ON)	320–335 mA	1.60–1.68 W	0.28–0.42 W	11–34 °C	25°C: 36–59°C 60°C: 71–94°C
5V peak (sum peaks)	600 mA	3.00 W	0.53–0.75 W	21–60 °C	25°C: 46–85°C 60°C: 81–120°C

Notes: LM2596 temperature rise strongly depends on PCB copper/heatsinking and the module implementation. Verify with a prototype thermal measurement at worst-case load.

3.2 AMS1117-3.3V (LDO) – 5V→3.3V

Formula: Ploss = (Vin − Vout) × Iout = 1.7V × Iout

Case	Iout	Pout	Ploss	ΔT est. (θJA 50–100)	Tj est. @ Ta=25/60
Average (actual 3.3V loads)	36 mA	0.119 W	0.061 W	3–6 °C	25°C: 28–31°C 60°C: 63–

					66°C
Peak (sum peaks on 3.3V)	73 mA	0.241 W	0.124 W	6–12 °C	25°C: 31–37°C 60°C: 66–72°C
Budget (design margin)	80 mA	0.264 W	0.136 W	7–14 °C	25°C: 32–39°C 60°C: 67–74°C

Conclusion: AMS1117 at ~36 mA average has very low thermal dissipation. The main thermal attention is on LM2596 (and the MQ-2 heater as a local heat source).